

Tuesday Warm-up: Electrostatics

Prof. Jordan C. Hanson

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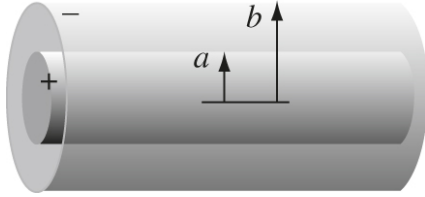


Figure 1: A positively charged cylinder with charge density $+\rho$, and a negatively charged cylindrical shell with charge density $-\sigma$.

1 Memory Bank

- Definition of electric field: $\mathbf{F} = q\mathbf{E}$.
- Permittivity of free space: $\epsilon_0 = 8.85 \times 10^{-12} \text{ N}^{-1} \text{ C}^2 \text{ m}^{-2}$
- Let $\rho(\mathbf{r}')$ represent the charge density.
- Let $Q_{\text{enc}} = \int \rho d\tau'$, the total charge within a volume.
- Coulomb's Law, discrete

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i \quad (1)$$

- Coulomb's Law, continuous

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}} d\tau' \quad (2)$$

- Gauss' Law, integral form

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}} \quad (3)$$

- Gauss' Law, differential form

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho \quad (4)$$

- Definition of potential

$$V(\mathbf{r}) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{l}' \quad (5)$$

- Relationship between potential and field:

$$\mathbf{E} = -\nabla V \quad (6)$$

2 Gauss' Law

1. A long coaxial cable (Fig. 1) carries a uniform volume charge density $+\rho$ on the inner cylinder (radius a), and a uniform surface charge density $-\sigma$ on the outer cylindrical shell (radius b). The cable as a whole is electrically neutral. Find the electric field in each of the following three regions: (i) inside the inner cylinder, (ii) between the cylinders, and (iii) outside the cable. Plot $|\mathbf{E}|$ as a function of r .

3 The Curl of E

1. Using Stoke's Theorem, show that the curl of the following vector field is zero:

$$\mathbf{v} = \frac{\hat{r}}{r^2} \quad (7)$$

2. Show that the potential of a point charge q at the origin is

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (8)$$